

Lab sheet 08**Submission Due: 08.07.2024****Title:** PHP, MySQL and JavaScript.**Aims:**

- Create and read data with PHP and MySQL
- Creating a simple web application using PHP, MySQL and JavaScript

Tasks:

- Create new database
- Implement functions for store data
- Implement functions for read specific user data
- Save sessions when user login

Activities :

1. Create a new database as “**expense_tracker**”
2. Create a new table named “**users**”

```
CREATE DATABASE expense_tracker;

USE expense_tracker;

CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  email VARCHAR(100) NOT NULL UNIQUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

3. Implement functions for user registration
I. JavaScript

```
async function register() {
  const username =
document.getElementById('registerUsername').value;
  const password =
document.getElementById('registerPassword').value;
  const email =
document.getElementById('registerEmail').value;

  const response = await fetch('php/register.php', {
```

```

        method: 'POST',
        body: JSON.stringify({ username, password, email
    })),
        headers: { 'Content-Type': 'application/json' }
    });
    const result = await response.json();
    alert(result.message);
}

```

II. register.php

```

<?php
    session_start();
    require_once '../database/User.php';

    $database = new Database("localhost", "root", "",
"expense_tracker");
    $db = $database->getConnection();

    $user = new User($db);

    $data = json_decode(file_get_contents("php://input"));

    $user->setUsername($data->username);
    $user->setPassword($data->password);
    $user->setEmail($data->email);

    if ($user->register()) {
        echo json_encode(["message" => "User was registered."]);
    } else {
        echo json_encode(["message" => "Unable to register the
user."]);
    }
?>

```

4. Implement functions for user login

I. JavaScript

```

async function login() {
    const username =
document.getElementById('loginUsername').value;
    const password =
document.getElementById('loginPassword').value;

    const response = await fetch('php/login.php', {
        method: 'POST',
        body: JSON.stringify({ username, password }),
    });
}

```

```
        headers: { 'Content-Type': 'application/json' }
    });
    const result = await response.json();
    if (result.message === "Login successful.") {
        window.location.href = 'dashboard.html';
    } else {
        alert(result.message);
    }
}
```

II. login.php

```
<?php
    session_start();
    require_once '../database/User.php';

    $database = new Database("localhost", "root", "",
"expense_tracker");
    $db = $database->getConnection();

    $user = new User($db);

    $data = json_decode(file_get_contents("php://input"));

    $user->setUsername($data->username);
    $user->setPassword($data->password);

    if ($user->login()) {
        $_SESSION['username'] = $user->username;
        echo json_encode(["message" => "Login successful."]);
    } else {
        echo json_encode(["message" => "Login failed."]);
    }
?>
```

5. Implement function for logout

I. JavaScript

```
async function logout() {
    const response = await fetch('php/logout.php', {
        method: 'POST'
    });
    const result = await response.json();
    if (result.message === "Logged out successfully.") {
        window.location.href = 'index.html';
    } else {
        alert(result.message);
    }
}
```

```
}  
}
```

II. logout.php

```
<?php  
    session_start();  
    session_destroy();  
    echo json_encode(["message" => "Logged out successfully."]);  
?>
```

Discussion:

- What is JSON ?
- Why use JSON ?